Self-similar counterexamples to the consecutive-degree conjectures of Lorentz

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Abstract

We construct two continuous functions on $[-1,1]$. The first vanishes at zero, is not odd, and has identical best uniform polynomial approximants at every consecutive pair of degrees $(2k+1, 2k+2)$. The second is not even and has identical best approximants at every pair $(2k, 2k+1)$. These give counterexamples to Conjectures $3'$ and 4 in the formulation of Saff and Varga. Both functions are uniformly convergent self-similar series with contraction $-3/5$ and explicit algebraic polynomial forcing. Three and four finite alternation certificates, respectively, propagate to every degree. A quantitative stability argument proves the certificates on the entire interval with the explicit perturbation parameter 10^{-10}. The finite algebraic identities and inequalities are verified by independent programs using exact rational arithmetic. The new constructions are not asserted here to have been formalized in Lean.

1 The statements and the proof strategy

For $f \in C[-1,1]$, write $P_n f$ for its unique best uniform approximant among real polynomials of degree at most n, and let $E_n(f) = \|f - P_n f\|_\infty$. Throughout, T_k is the Chebyshev polynomial defined by $T_k(\cos\theta) = \cos(k\theta)$.

The two assertions considered here are the normalized oddness conjecture and the evenness conjecture from [1]:

$$\begin{aligned} f(0) = 0, \quad P_{2k+1} f = P_{2k+2} f \ (k \geq 0) \quad &\Longrightarrow \quad f \text{ is odd}, \\ P_{2k} f = P_{2k+1} f \ (k \geq 0) \quad &\Longrightarrow \quad f \text{ is even}. \end{aligned}$$

Theorem 1.1. *There exist $f_{\mathrm{o}}, f_{\mathrm{e}} \in C[-1,1]$ such that*

$$\begin{aligned} f_{\mathrm{o}}(0) = 0, \qquad P_{2k+1} f_{\mathrm{o}} = P_{2k+2} f_{\mathrm{o}} \quad (k \geq 0), \qquad & f_{\mathrm{o}}(1) + f_{\mathrm{o}}(-1) \neq 0, \\ P_{2k} f_{\mathrm{e}} = P_{2k+1} f_{\mathrm{e}} \quad (k \geq 0), \qquad & f_{\mathrm{e}}(1) - f_{\mathrm{e}}(-1) \neq 0. \end{aligned}$$

Both functions are specified below by algebraic polynomial forcing and the fixed parameter $\varepsilon = 10^{-10}$.

The proof has three dependencies. First, Chebyshev composition reduces all degree conditions to finitely many initial ones. Second, an explicit baseline has known extremal points and sufficiently strong local gaps to preserve finite contact inequalities under perturbation. Third, exact algebraic certificates supply the required initial approximants. We prove these ingredients in that order and then assemble the two constructions.

2 Self-similarity and the finite degree reduction

Set

$$a = \frac{3}{5}, \qquad u = \frac{\sqrt{2}}{2}, \qquad \mathcal{S}h(x) = \sum_{j=0}^{\infty} (-a)^j h(T_{3^j}(x)), \qquad G = \mathcal{S}T_1.$$

For continuous h, the series converges uniformly on $[-1,1]$, since $\|h \circ T_{3^j}\|_\infty = \|h\|_\infty$ and $\sum a^j < \infty$. In particular, $\mathcal{S}h$ is continuous and

$$\mathcal{S}h = h - a(\mathcal{S}h) \circ T_3. \tag{1}$$

Commutation of Chebyshev polynomials gives $\mathcal{S}T_d = G \circ T_d$.

Lemma 2.1 (Chebyshev composition)**.** *For $F \in C[-1,1]$, $m \geq 1$, and $n \geq 0$,*

$$P_n(F \circ T_m) = (P_{\lfloor n/m \rfloor} F) \circ T_m.$$

Proof. Write $x = \cos\theta$. Average any polynomial competitor $p(\cos\theta)$ over the shifts $\theta \mapsto \theta + 2\pi j/m$, $0 \leq j < m$. The target $F(\cos m\theta)$ is invariant under these shifts, so this averaging cannot increase the uniform error. Only cosine frequencies divisible by m survive. The average is therefore $r(\cos m\theta)$ for a polynomial r of degree at most $\lfloor n/m \rfloor$. Since T_m maps $[-1,1]$ onto itself, its error equals $\|F - r\|_\infty$. This gives equality of the two best errors, and uniqueness gives the claimed identity. □

Lemma 2.2 (Recurrence for best approximants)**.** *If q is a polynomial and $f = \mathcal{S}q$, then*

$$P_n f = q - a(P_{\lfloor n/3 \rfloor} f) \circ T_3, \qquad n \geq \deg q. \tag{2}$$

Proof. For $n \geq \deg q$, translation by q preserves the space of degree-at-most n polynomials. Apply this translation, homogeneity of best approximation, (1), and Lemma 2.1. □

Proposition 2.3 (Finite conditions for the two parity patterns)**.** *Let $f = \mathcal{S}q$.*

1. *If $\deg q \leq 7$ and $P_1 f = P_2 f$, $P_3 f = P_4 f$, and $P_5 f = P_6 f$, then $P_{2k+1} f = P_{2k+2} f$ for every $k \geq 0$.*
2. *If $\deg q \leq 8$ and $P_0 f = P_1 f$, $P_2 f = P_3 f$, $P_4 f = P_5 f$, and $P_6 f = P_7 f$, then $P_{2k} f = P_{2k+1} f$ for every $k \geq 0$.*

Proof. For the first claim, use strong induction on odd n. The cases $n = 1, 3, 5$ are assumptions. For odd $n \geq 7$, the two floor indices in (2) for $n, n+1$ agree unless $n = 6r + 5$. In that case they are $2r + 1, 2r + 2$, a smaller odd/even pair covered by induction.

For the second claim, use strong induction on even n. The cases $n = 0, 2, 4, 6$ are assumptions. For even $n \geq 8$, the floor indices agree unless $n = 6r + 2$, in which case they are the smaller even/odd pair $2r, 2r + 1$. The same recurrence completes the proof. □

3 The baseline and its exact extrema

For $0 < c < 1$, define

$$G_c(x) = \sum_{j \geq 0} (-c)^j T_{3^j}(x).$$

Thus $G = G_a$. Every G_c is odd. We first establish the extremal geometry for $b = 1/3$ and then transfer it to $a = 3/5$.

Lemma 3.1. *The function G has unique maximum at u, unique minimum at $-u$, and*

$$B := G(u) = \|G\|_\infty = \frac{u}{1-a} = \frac{5u}{2}. \tag{3}$$

Moreover, for every $y \in [-1,1]$,

$$B - |G(y)| \geq \frac{1}{2}\operatorname{dist}(y, \{-u, u\})^2. \tag{4}$$

Proof. Pair adjacent terms in G_b:

$$G_b = \sum_{j\geq 0} 9^{-j} g \circ T_{9^j}, \qquad g(x) = x - \frac{1}{3}T_3(x) = 2x - \frac{4}{3}x^3.$$

The identity

$$g(u) - g(y) = \frac{4}{3}(y-u)^2(y+2u)$$

shows that g has its unique maximum at u and its unique minimum at $-u$ on the interval. Since $T_9(u) = u$, every term in the paired series has its maximum at u. Consequently $B_b := \|G_b\|_\infty = 3u/2$ and

$$B_b - G_b(y) \geq \frac{4}{3}(y-u)^2(y+2u) \geq \frac{1}{2}(y-u)^2. \tag{5}$$

The last inequality uses $\frac{4}{3}(\sqrt{2}-1) > 1/2$. Oddness gives the corresponding lower bound for $B_b + G_b(y)$.

The uniformly convergent series satisfy the identity

$$G_a(x) = G_b(x) + (a-b)\sum_{j=1}^{\infty} a^{j-1}(-1)^j G_b(T_{3^j}(x)). \tag{6}$$

Indeed, after expansion, the coefficient of T_{3^k} on the right is

$$(-1)^k \left[b^k + (a-b)\sum_{j=1}^{k} a^{j-1} b^{k-j} \right] = (-a)^k.$$

Rearrangement is justified by absolute uniform convergence.

Every coefficient multiplying a function on the right of (6) is positive. Each such function has norm B_b and value B_b at u, because $T_3(u) = -u$ and G_b is odd. Thus

$$G_a(u) = B_b\left(1 + \frac{a-b}{1-a}\right) = \frac{u}{1-a}.$$

All deficits from this maximum are nonnegative. Keeping the first deficit and using (5) proves $B - G_a(y) \geq (y-u)^2/2$. Oddness proves $B + G_a(y) \geq (y+u)^2/2$. Their minimum is (4), and also proves uniqueness of the two extrema. □

For later use, $P_0 G = 0$ by oddness. Lemma 2.2 applied to $G = \mathcal{S}T_1$ therefore gives

$$P_1 G = P_2 G = T_1. \tag{7}$$

It follows that $G \circ T_d$ has initial best approximant zero for $0 \leq n < d$, and best approximant T_d for $d \leq n < 3d$. The residuals in these two ranges are $G \circ T_d$ and $-aG \circ T_{3d}$.

4 From finite contact inequalities to the whole interval

The point of the larger contraction is that the baseline gap and an arbitrary polynomial-forcing perturbation can be compared at the same scale. No stationarity conditions on the perturbation are needed.

Lemma 4.1 (A gap at the exit scale)**.** *Let $0 < |s| \le \delta := 1/16$, and choose $J \ge 0$ minimal with $3^J|s| \ge \delta$. Then*

$$B - G(\cos(\pi/4 + s)) \ge \frac{a^J}{4608}. \tag{8}$$

Proof. Minimality gives $\delta \le 3^J|s| \le 3\delta$. The exact phase relation, using $3^J \equiv 1$ or $3 \pmod 8$ according to the parity of J, is

$$(-1)^J G_b(\cos(3^J(\pi/4 + s))) = G_b(\cos(\pi/4 + (-1)^J 3^J s)).$$

On $[\pi/4 - 3\delta, \pi/4 + 3\delta]$, sine is at least $1/2$. Thus the argument of the final G_b is at distance at least $\delta/2$ from u, and its deficit is at least $\delta^2/8 = 1/2048$ by (5). For $J \ge 1$ the coefficient of this term in (6) is $(a-b)a^{J-1} = (4/9)a^J$. For $J = 0$ the coefficient is $1 \ge 4/9$. Discarding all other nonnegative deficits proves (8). □

For a polynomial $w = \sum w_k T_k$, the bounds

$$\|w\|_\infty \le \sum |w_k|, \qquad \sup_\theta \left| \frac{d}{d\theta} w(\cos\theta) \right| \le \sum k|w_k|$$

follow directly from the cosine representation. The second bound is in the angular variable, not the spatial variable.

Lemma 4.2 (Perturbation oscillation)**.** *Let $H = \mathcal{S}h$ and $D = H - v$, where h, v are polynomials. Suppose*

$$\|h\|_\infty \le M, \quad \sup_\theta |(h(\cos\theta))'| \le L, \quad \|v\|_\infty \le V, \quad \sup_\theta |(v(\cos\theta))'| \le L_v.$$

Let $d \ge 1$ and let J be a nonnegative integer with $\delta \le 3^J d|t| \le 3\delta$. Then

$$|D(\cos(\theta + t)) - D(\cos\theta)| \le C_d a^J, \qquad C_d = 5M + \frac{15L}{64d} + \frac{3L_v}{16d}. \tag{9}$$

Also $\|D\|_\infty \le K := \frac{5}{2}M + V$.

Proof. The jth forcing difference is at most $\min\{L3^j|t|, 2M\}$. Splitting the series before J gives

$$\begin{aligned} |H(\cos(\theta + t)) - H(\cos\theta)| &\le L|t| \sum_{j<J} (3a)^j + 2M \sum_{j \ge J} a^j \\ &\le \left(\frac{15L}{64d} + 5M \right) a^J. \end{aligned}$$

Here $3a - 1 = 4/5$ and $3^J|t| \le 3/(16d)$. Moreover $3^{-J} \le a^J$, so $L_v|t| \le 3L_v a^J/(16d)$. This proves (9). The norm bound follows from $\|H\|_\infty \le M/(1-a) = 5M/2$. □

Proposition 4.3 (Quantitative finite-contact stability)**.** *Let*

$$R(x) = sAG(T_d(x)), \qquad A > 0, \quad s \in \{-1, 1\}, \quad d \geq 1.$$

Its extremal points are the $2d$ *zeros of* T_{2d}*, namely* $x_i = \cos((2i+1)\pi/(4d))$*,* $0 \leq i < 2d$*. Put* $\sigma_i = R(x_i)/(AB)$*. For* $D = \mathcal{S}h - v$*, suppose*

$$\sigma_i D(x_i) \leq \lambda \qquad (0 \leq i < 2d). \tag{10}$$

With the constants of Lemma 4.2, set

$$\eta_0 = \frac{1}{1296}, \qquad \gamma = \frac{1}{8 \cdot 1296^2}.$$

If $\varepsilon > 0$ *satisfies*

$$\varepsilon C_d \leq \frac{A}{4608}, \qquad \varepsilon(K + |\lambda|) \leq A\gamma, \tag{11}$$

then

$$|R(x) + \varepsilon D(x)| \leq AB + \varepsilon\lambda \qquad (x \in [-1, 1]). \tag{12}$$

Every equality in (10) *is an equality in* (12) *with sign* σ_i*.*

Proof. Use $x = \cos\theta$. Around every extremal angle θ_i take the neighborhood $|\theta - \theta_i| \leq \eta_0/d$. By periodicity, reflection, and oddness, the signed baseline value reduces to $AG(\cos(\pi/4 + s'))$, with $|s'| = d|\theta - \theta_i|$. For $s' \neq 0$ choose its exit scale J. Lemmas 4.1 and 4.2 give

$$\sigma_i(R(x) + \varepsilon D(x)) \leq AB - \frac{Aa^J}{4608} + \varepsilon\lambda + \varepsilon C_d a^J \leq AB + \varepsilon\lambda.$$

At the contact itself this follows from (10).

We must also control the opposite sign. The neighborhood radius ensures $J \geq 4$. Apply the forcing estimate in Lemma 4.2 to $G = \mathcal{S}T_1$, taking $d = 1$, $M = L = 1$, and omitting v. Then

$$|G(\cos(\pi/4 + s')) - B| \leq \frac{335}{64}a^J \leq \frac{335}{64}\left(\frac{3}{5}\right)^4 < \frac{679}{1000} < \frac{B}{2}.$$

Thus $\sigma_i R(x) > AB/2$. The second inequality in (11) gives $\varepsilon \|D\|_\infty \leq A\gamma < AB/2$. The perturbed residual retains this sign, proving the absolute bound throughout the neighborhoods. At $s' = 0$, sign retention follows directly from $AB - \varepsilon K > 0$.

Outside these neighborhoods, the angular distance of $d\theta$ from every preimage of $\pm u$ is at least η_0. Monotonicity of cosine between its extrema, and the sine bound $1/2$ in a short interval adjacent to each preimage, give $\operatorname{dist}(\cos(d\theta), \{-u, u\}) \geq \eta_0/2$. By (4), $|R(x)| \leq AB - A\gamma$. Therefore

$$|R(x) + \varepsilon D(x)| \leq AB - A\gamma + \varepsilon K \leq AB + \varepsilon\lambda.$$

Finally, $AB + \varepsilon\lambda > 0$ because $\varepsilon|\lambda| \leq A\gamma < AB$. At an active contact the signed value is exactly $AB + \varepsilon\lambda$, as claimed. □

In both constructions below the exact certificates give

$$M < 32, \quad L < 128, \quad V < 32, \quad L_v < 128, \quad |\lambda| < 32. \tag{13}$$

All residuals have $A \geq 3/5$ and $d \geq 3$. Thus $C_d < 178$ and $K + |\lambda| < 144$. The single explicit choice

$$\varepsilon = 10^{-10} \tag{14}$$

satisfies (11), since

$$\frac{178}{10^{10}} < \frac{3}{5 \cdot 4608}, \qquad \frac{144}{10^{10}} < \frac{3}{5 \cdot 8 \cdot 1296^2}.$$

Consequently, any of the finite certificates below with the stated bounds is automatically a certificate on the whole interval.

5 The normalized oddness counterexample

Define

$$h_{\mathrm{o}} = (\eta - \sqrt{3} - 1)T_0 - \sqrt{3}\,T_2 + T_4 - 3T_5 + \eta T_6 + \beta T_7, \tag{15}$$

$$\eta = \frac{16\cos(\pi/18) - 26\cos(\pi/6) - 26\cos(5\pi/18)}{15}. \tag{16}$$

To specify β exactly, put $c_j = \cos(j\pi/36)$ and set

$$\begin{aligned} 7132971\beta = {} & -21647763 + 7264892c_1 - 16364700c_2 + 2333622c_3 - 20710080c_4 \\ & + 7960924c_5 - 8377290c_6 + 6354480c_7 - 6380100c_8 \\ & + 8032814c_9 - 667710c_{10} + 24678c_{11}. \end{aligned} \tag{17}$$

Numerically $\eta \approx -1.564814286803340$ and $\beta \approx -5.914876184323304$. These decimals are descriptive only; (16)–(17) define the coefficients.

Set

$$f_{\mathrm{o}} = G \circ T_3 + \varepsilon \mathcal{S} h_{\mathrm{o}} = \mathcal{S}(T_3 + \varepsilon h_{\mathrm{o}}). \tag{18}$$

We next give the three finite certificates needed for this function.

5.1 Exact definition of the correction polynomials

Let

$$x_j = -\cos\frac{(2j+1)\pi}{36}, \quad 0 \le j < 18, \qquad \sigma_j = (1,-1,-1,1)_{j \bmod 4}.$$

The coarse grid is $y_i = x_{3i+1}$, $0 \le i < 6$, with signs $\tau = (-1,1,1,-1,-1,1)$. These are respectively the extrema of $-aG \circ T_9$ and $G \circ T_3$.

Define the explicit grid functions

$$W_k(x_j) = T_k(x_j) - aT_{3k}(x_j) + \frac{a^2}{1 + (-1)^k a} T_{9k}(x_j).$$

They equal $(\mathcal{S}T_k)(x_j)$: after two iterates the finite orbit is on the two-cycle $\{\pm u\}$, and the remaining series is geometric. Equivalently, the exact identities $W_k(x_j) + aW_k(T_3 x_j) = T_k(x_j)$ determine these values uniquely. Write $H_j = \sum_{k=0}^{7} (h_{\mathrm{o}})_k W_k(x_j)$.

The correction polynomials v_1, v_3, v_5 and scalars $\lambda_2, \lambda_4, \lambda_6$ are specified by the contact equations

$$\begin{aligned} v_1(y_i) + \tau_i \lambda_2 &= H_{3i+1}, && i \in I_2 := \{0,1,4,5\}, \\ v_3(x_j) + \sigma_j \lambda_4 &= H_j, && j \in I_4 := \{0,2,4,13,15,17\}, \\ v_5(x_j) + \sigma_j \lambda_6 &= H_j, && j \in I_6 := \{0,2,4,6,11,13,16,17\}, \end{aligned}$$

with degrees at most $1, 3, 5$, respectively. The systems are consistent for the exact values (16)–(17). Each has full column rank, so the correction and scalar are unique. Their exact rational coordinates in the field described below are included in `odd_certificate.json`.

For orientation, their Chebyshev coefficients are:

coefficient	v_1	v_3	v_5
T_0	−3.310540684	−3.310540684	−3.380936697
T_1	−0.781038720	0.805215314	−1.373864474
T_2	0	−1.138509891	−0.999856812
T_3	0	−2.959735280	1.640309342
T_4	0	0	−0.070396013
T_5	0	0	−4.379580520

The corresponding scalars are approximately 4.600544015619, 7.815043886301, and 6.118310129767.

Lemma 5.1 (Odd-pattern finite certificate)**.** *The above corrections satisfy the bounds* (13)*. At every coarse or fine node, as appropriate,*

$$\tau_i(H_{3i+1} - v_1(y_i)) \le \lambda_2, \qquad \sigma_j(H_j - v_{n-1}(x_j)) \le \lambda_n \quad (n = 4, 6).$$

Equality holds exactly on I_2, I_4, I_6, respectively. Each active set has alternating signs. Every inactive slack is greater than $3/50$*.*

Exact arithmetic verification. Put $z = 2\cos(\pi/36)$. All entries lie in

$$\mathbb{Q}[z], \qquad M(z) = z^{12} - 12z^{10} + 54z^8 - 112z^6 + 105z^4 - 36z^2 + 1 = 0.$$

The relevant root is isolated in $(1.98, 2)$. To see uniqueness there, write $M = S_6^2 - 3$, where $S_6 = z^6 - 6z^4 + 9z^2 - 2$; on that interval $S_6 > 0$ and $S_6' = 6z(z^2 - 1)(z^2 - 3) > 0$.

The program `verify_odd.py` constructs every coefficient by exact field arithmetic, checks the contact equations, and checks the displayed closed expressions for η, β. It encloses the root by 250 rational bisections and evaluates every nonzero slack by rational interval arithmetic. The lower bounds exceed $3/50$; the least positive slacks for the three certificates exceed 1.5265, 1.6678, and 0.0619, respectively. Exact interval bounds also establish (13), $\eta < 0$, and positivity of the three scalars.

The independently implemented `audit_odd.py` starts from the serialized rational coefficients. It checks the root isolation, ordered nodes, cubic node map, finite functional equation for the actual H, all slack identities, their signs, and the alternating active sets. It also checks (11) directly for $\varepsilon = 10^{-10}$. These programs use rational arithmetic for every acceptance condition; floating-point conversion is used only in printed summaries. □

5.2 Completion of the first construction

For degrees $1, 2$, the baseline approximant is zero, so the residual for the candidate εv_1 is $G \circ T_3 + \varepsilon(\mathcal{S}h_{\mathrm{o}} - v_1)$. For degrees $3, 4$ and $5, 6$, the baseline approximant is T_3, and the residual is $-aG \circ T_9 + \varepsilon(\mathcal{S}h_{\mathrm{o}} - v)$. Apply Proposition 4.3 with $(A, s, d) = (1, 1, 3)$ or $(a, -1, 9)$. The finite certificate provides respectively four, six, and eight alternating contacts, while the candidates have degrees at most $1, 3, 5$. Chebyshev alternation therefore gives

n	$P_n f_{\mathrm{o}}$	$E_n(f_{\mathrm{o}})$
$1, 2$	εv_1	$B + \varepsilon\lambda_2$
$3, 4$	$T_3 + \varepsilon v_3$	$aB + \varepsilon\lambda_4$
$5, 6$	$T_3 + \varepsilon v_5$	$aB + \varepsilon\lambda_6$.

Proposition 2.3 now proves every odd/even plateau.

The identity $h_{\mathrm{o}}(0) = 0$ follows directly from $T_{2j}(0) = (-1)^j$. Thus (1) at zero gives $f_{\mathrm{o}}(0) = 0$. At the endpoints,

$$h_{\mathrm{o}}(1) + h_{\mathrm{o}}(-1) = 4(\eta - \sqrt{3}).$$

Since $T_3(\pm 1) = \pm 1$ and $G \circ T_3$ is odd,

$$f_{\mathrm{o}}(1) + f_{\mathrm{o}}(-1) = \frac{4\varepsilon(\eta - \sqrt{3})}{1 + a} = \frac{5}{2}\varepsilon(\eta - \sqrt{3}) \neq 0. \tag{19}$$

This proves the first assertion of Theorem 1.1.

6 The evenness counterexample

This construction uses the baseline $G \circ T_6$ and a degree-eight forcing perturbation. We specify its coefficients by a small, explicit linear system, avoiding long rational coordinate lists in the text.

6.1 Exact algebraic definition

The unknown forcing is

$$h_{\mathrm{e}} = c_1T_1 + c_2T_2 + c_3T_3 + c_4T_4 + c_5T_5 + c_7T_7 + c_8T_8.$$

Thus $c_0 = c_6 = 0$. Introduce correction polynomials w_0, w_2, w_4, w_6 of degrees at most $0, 2, 4, 6$, and scalars $\mu_1, \mu_3, \mu_5, \mu_7$.

For $m = 12, 36$, define the increasing nodes

$$x_j^{(m)} = -\cos\frac{(2j+1)\pi}{2m}, \qquad 0 \le j < m.$$

Use $m_n = 12$ for $n = 1, 3, 5$ and $m_7 = 36$. Let

$$\sigma_{n,j} = \begin{cases} (1,-1,-1,1)_{j \bmod 4}, & n = 1,3,5, \\ (-1,1,1,-1)_{j \bmod 4}, & n = 7. \end{cases}$$

For a grid angle θ set $J = 1$ when $m = 12$ and $J = 2$ when $m = 36$, and define

$$W_k^{(m)}(\cos\theta) = \sum_{i=0}^{J-1}(-a)^i\cos(k3^i\theta) + \frac{\sum_{i=J}^{J+3}(-a)^i\cos(k3^i\theta)}{1-a^4}. \tag{20}$$

After J iterates these node orbits have period dividing four. Therefore (20) is the exact value of $\mathcal{S}T_k$ at the node.

Write $\mathcal{H}_j^{(m)} = \sum_k c_k W_k^{(m)}(x_j^{(m)})$. The defining equations are

$$w_{n-1}(x_j^{(m_n)}) + \sigma_{n,j}\mu_n = \mathcal{H}_j^{(m_n)}, \qquad j \in J_n, \tag{21}$$

where

$$\begin{aligned} J_1 &= \{3, 6, 8\}, \\ J_3 &= \{1, 3, 4, 6, 8, 10\}, \\ J_5 &= \{0, 2, 4, 6, 7, 9, 11\}, \\ J_7 &= \{3, 6, 8, 13, 19, 22, 27, 30, 32\}. \end{aligned}$$

Add the two normalizations

$$[T_2]w_2 = 1, \qquad [T_4]w_4 = -1. \tag{22}$$

Here $[T_k]w$ denotes the coefficient in the Chebyshev basis. There are 27 equations for the $7 + 1 + 3 + 5 + 7 + 4 = 27$ unknowns. Exact elimination shows that this system is nonsingular. Equations (20)–(22) therefore define the forcing and corrections unambiguously by real algebraic numbers.

Their approximate forcing coefficients are

k	c_k
1	-0.017561786853260
2	0.286725254171375
3	0.018613141564351
4	-0.862027229474975
5	-0.013885677377745
7	0.026299792766218
8	$-0.253316709984462.$

The exact rational coordinate lists are in `even_certificate.json`. Define

$$f_{\mathrm{e}} = G \circ T_6 + \varepsilon \mathcal{S} h_{\mathrm{e}} = \mathcal{S}(T_6 + \varepsilon h_{\mathrm{e}}). \tag{23}$$

Lemma 6.1 (Even-pattern finite certificate)**.** *For each $n = 1, 3, 5, 7$ and every point of the corresponding grid,*

$$\sigma_{n,j}\big(\mathcal{H}_j^{(m_n)} - w_{n-1}(x_j^{(m_n)})\big) \le \mu_n.$$

The equality sets are exactly J_n. Alternating subsets of cardinality $n+2$ are J_1, J_5, J_7, and $\{1, 3, 6, 8, 10\} \subset J_3$. All nonzero slacks exceed $1/50$. The coefficient bounds (13) *hold, and*

$$c_1 + c_3 + c_5 + c_7 > \frac{1}{100}. \tag{24}$$

Exact arithmetic verification. All coefficients belong to the real algebraic field generated by $z = 2\cos(\pi/72)$, whose defining polynomial is

$$\begin{aligned} M_{24}(z) = {} & z^{24} - 24z^{22} + 252z^{20} - 1520z^{18} + 5814z^{16} - 14688z^{14} \\ & + 24752z^{12} - 27456z^{10} + 19305z^8 - 8008z^6 + 1716z^4 - 144z^2 + 1. \end{aligned}$$

It satisfies $M_{24}(z) = M(z^2 - 2)$, where M is the polynomial used in the odd certificate. The relevant root is isolated in $(1.998, 2)$; the same monotonicity argument, applied to $z^2 - 2$, proves uniqueness.

The generator `generate_even.py` solves the 27-by-27 system by exact arithmetic, with a nonzero pivot at every step. It checks every functional identity, equality, and slack and encloses the root by 400 rational bisections. The minimum positive slacks for $n = 1, 3, 5, 7$ exceed 0.1336, 0.0936, 0.0965, and 0.0230, respectively. Exact coefficient bounds give $M < 2$, $L < 34$, $V < 2$, $L_v < 72$, and $|\mu_n| < 1$, which imply (13); it also verifies (24).

The independent standard-library program `audit_even.py` uses a separate implementation of rational polynomial arithmetic. From the serialized coefficients it reconstructs all 36 fine nodes, verifies their ordering and cubic map, detects the finite orbit cycles, and evaluates the exact orbit sums. It then verifies every slack, equality, alternating set, coefficient bound, and (11). No floating-point tolerance is used to accept any identity or inequality. □

For reference, the scalar error corrections are approximately

$$\mu_1 = 0.577201789619347, \qquad \mu_3 = 0.223648399026073,$$
$$\mu_5 = -0.129582723095644, \qquad \mu_7 = 0.276968241048616.$$

The negative value of μ_5 causes no difficulty: the stability proposition uses $|\mu_5|$ in its smallness condition, and the total error remains positive.

6.2 Completion of the second construction

By (7) and Chebyshev composition, the baseline $G \circ T_6$ has best approximant zero for degrees 0 through 5 and best approximant T_6 for degrees $6, 7$. Its residuals are consequently $G \circ T_6$ and $-aG \circ T_{18}$.

Apply Proposition 4.3 with $(A, s, d) = (1, 1, 6)$ for the first three certificates and $(a, -1, 18)$ for the fourth. Alternation gives

n	$P_n f_{\mathrm{e}}$	$E_n(f_{\mathrm{e}})$
$0, 1$	εw_0	$B + \varepsilon\mu_1$
$2, 3$	εw_2	$B + \varepsilon\mu_3$
$4, 5$	εw_4	$B + \varepsilon\mu_5$
$6, 7$	$T_6 + \varepsilon w_6$	$aB + \varepsilon\mu_7$.

Since $\deg(T_6 + \varepsilon h_{\mathrm{e}}) \le 8$, Proposition 2.3 extends these equalities to every even/odd pair.

Finally, $G \circ T_6$ is even, while the endpoint difference of the perturbation is

$$f_{\mathrm{e}}(1) - f_{\mathrm{e}}(-1) = \frac{2\varepsilon}{1+a}(c_1 + c_3 + c_5 + c_7) > \frac{\varepsilon}{80} > 0. \tag{25}$$

This proves the second assertion of Theorem 1.1.

7 Verification, scope, and reproducibility

The proof consists of the analytic estimates above and finite exact algebraic certificates. The latter are not numerical Remez calculations: all accepted identities are polynomial identities modulo a defining polynomial, and all accepted strict inequalities are certified by rational intervals at a uniquely isolated real root. The independently implemented audits also check that the finite orbit values are the values of the actual infinite series and that the explicit perturbation size satisfies the whole-interval estimates.

From the accompanying directory, run

```
python3 audit_odd.py
python3 audit_even.py
```

Both audits use only the Python standard library and the supplied JSON certificates. To regenerate the odd certificate, run `python3 verify_odd.py`. The optional even generator uses `python-flint`, installed by `python3 -m pip install python-flint`, and is run with `python3 generate_even.py`. Its output is independently audited without that dependency.

These constructions are distinct from the earlier counterexample to the common-origin-zero conjecture. Their best approximants are required to have the indicated degree plateaus; they are not required to vanish at zero. The first function itself is normalized at zero. The preceding Lean formalization of the common-origin-zero example is not a Lean formalization of the two new results here. The new proof status is a full analytic argument with two independently checked exact arithmetic certificates.

References

[1] E. B. Saff and R. S. Varga, *Remarks on some conjectures of G. G. Lorentz*, Journal of Approximation Theory **30** (1980), 29–36. https://www.math.kent.edu/~varga/pub/paper_126.pdf.